

SAR image former — detailed design

The as-built design of the spotlight-mode SAR image-formation processor on PolarFire SoC MPFS250T_ES. This is the reference for *how the thing actually works*: dataflow, numeric contracts, memory map, coherency rules, control interface, and the on-board data path.

Companion documents:

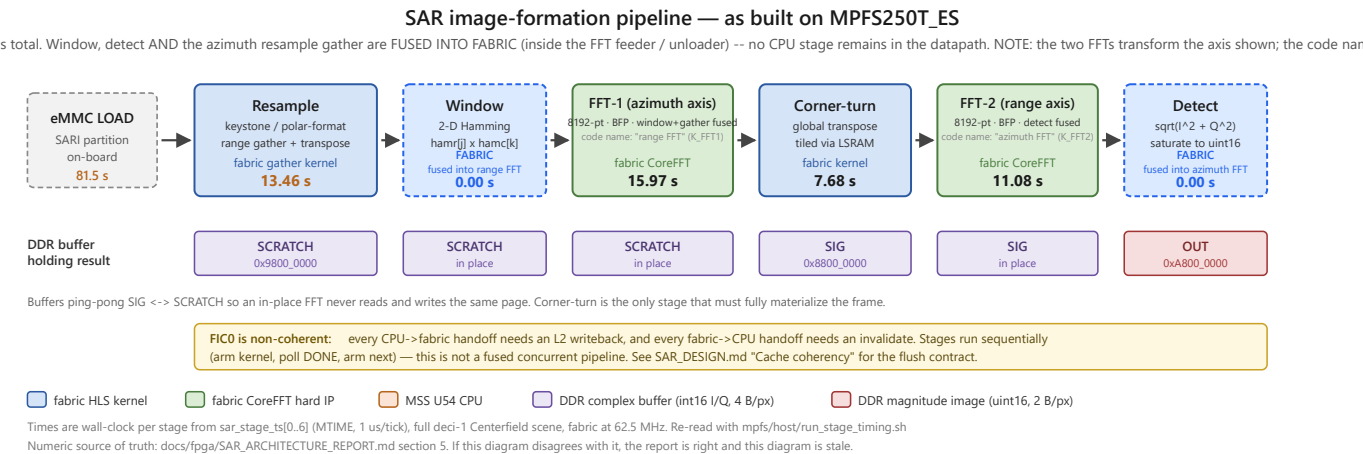
Document	Covers
fpga/SAR_ARCHITECTURE_REPORT.md	Measured per-stage timing and fabric resource usage — the numeric source of truth
fpga/SAR_PIPELINE_STATUS.md	Status, engine history, latency roadmap
fpga/SAR_PIPELINE_PROCESS.md	The PFA math and its host-side golden cross-reference
fpga/SILICON_ISO_TEST_RUNBOOK.md	JTAG bring-up and debug procedure — read before any silicon debug
PROJECT_SOURCE_OF_TRUTH.md	Authoritative index and anti-hallucination rules

1. What it computes

Input is an Umbra CPHD phase-history array (spotlight, X-band, single channel, complex float32). Output is a focused magnitude image. The algorithm is the Polar-Format Algorithm: interpolate the polar-sampled phase history onto a Cartesian k-space grid (the keystone resample), taper it, then take a separable 2-D FFT and detect.

The processing frame is a fixed 8192 x 8192 complex grid. A full capture is decimated to fit; per-axis sizing and rejection rules live in `mpfs/host/ddr_layout.py` (`plan_frame` / `check_input_dims`). Only the native 8192-point transform exists — 16384-point and multi-length paths were dropped.

2. Pipeline



Stages run sequentially: the MSS arms a kernel, polls its DONE flag, then arms the next. This is not a fused concurrent pipeline. Every stage is a DDR-to-DDR streaming pass, because the frame (256 MiB complex) far exceeds on-chip SRAM; on-chip each stage holds only a row, a transpose tile, or AXI burst FIFOs.

#	Stage	Engine	In	Out
1	Resample (range + transpose + azimuth)	fabric gather kernel + MSS coefficients	SIG	SCRATCH
2	Window (2-D Hamming)	<i>fused into the range-FFT feeder</i>	—	—
3	Range FFT	fabric CoreFFT	SCRATCH	SCRATCH
4	Corner-turn (transpose)	fabric kernel	SCRATCH	SIG
5	Azimuth FFT	fabric CoreFFT	SIG	SIG
6	Detect (magnitude)	<i>fused into the azimuth-FFT unloader (fabric)</i>	—	—

Total 48.19 s (2026-07-22, azimuth-gather fused); the per-stage breakdown is in [fpga/SAR_ARCHITECTURE_REPORT.md §5](#), which is the single numeric source of truth. Orchestration is `sar_form_image()` in `src/sar/sar_sequencer.c`.

2.0 Kernel-level decomposition

The table above is the *timing* view: six instrumented stages matching `sar_stage_ts[0..6]`. At kernel granularity there are now seven arm/wait cycles (the window kernel is no longer armed), because the resample stage internally runs range-resample, a corner-turn, and azimuth-resample before reporting one timestamp. Both views are correct; this is the one to use when reasoning about buffer traffic or arming order.

#	Stage	Kernel	Reads	Writes	Timing stage
1	range resample (per pulse, xM)	RES	SIG + COEF idx/wq	SCRATCH (row <code>invorder[i]</code> , tan φ -sorted)	resample
2	corner-turn (transpose)	CT	SCRATCH	SIG	resample
3	azimuth resample (per range bin, xNp)	RES	SIG + COEF	SCRATCH (uniform k-space)	resample
4	window (2-D Hamming)	<i>fused into FEED</i>	—	—	window (now 0.00 s)
5	range FFT	FEED -> CoreFFT -> <code>fft_unloader</code>	SCRATCH (stream)	SCRATCH (AXI4 write)	rangeFFT
6	corner-turn (transpose)	CT	SCRATCH	SIG	cornerturn

#	Stage	Kernel	Reads	Writes	Timing stage
7	azimuth FFT	FEED -> CoreFFT -> <code>fft_unloader</code>	SIG (stream)	SIG (AXI4 write)	azFFT
8	detect (magnitude)	fused into UNLD	—	OUT (final image)	detect (now 0.00 s)

Note that the corner-turn appears twice — once inside resample and once between the two FFT passes — and that steps 2 and 6 both write SIG while steps 1 and 3 both write SCRATCH. This is the SIG/SCRATCH ping-pong; a stage that reloads a buffer it just wrote is a bug, not an optimisation.

2.1 Resample (keystone / polar-format interpolation)

Two passes with a transpose between them, all inside `resample_2pass()`.

- Pass 1 (range): each real pulse row of SIG (N samples) is resampled to the padded width N_p and written to SCRATCH at its `tan_phi`-sorted row `invord[i]`, so SCRATCH ends up pulse-sorted. Padded rows $M..M_p-1$ are then zeroed.
- Transpose SCRATCH -> SIG so range bins become rows.
- Pass 2 (azimuth): each range-bin row (M sorted pulses) is resampled to M_p , leaving the resampled k-space in SCRATCH.

The interpolation contract is a two-tap linear gather:

```
out[i] = in[idx[i]] + (in[idx[i]+1] - in[idx[i]]) * wq[i] / 32768
```

equivalently `in[idx]·(1-w) + in[idx+1]·w` with $w = wq/2^{15}$. `idx` is an index into the source in its natural order; `idx = -1` means out-of-range and zero-fills.

Coefficients are computed just-in-time on the MSS, one line at a time, from the small per-pulse geometry arrays — precomputing the full grid would be ~768 MiB. The MSS double-buffers: while the fabric kernel gathers line `i` from bank `b`, the CPU fills bank `b+1` for line `i+1`. Coefficient generation is float on the CPU (`sar_resample_coeffs.c`, mirrors the host `interp_coeffs()` at corr 1.0); the interpolation itself is fixed-point in fabric.

Cache maintenance on this per-line path used to dominate the stage. Replacing the whole-L2 `flush_l2_cache()` with the targeted coefficient-bank writeback of section 6 removed 13,826 whole-L2 flushes and took the stage from 53.6 s to 29.2 s — 24.4 s, or about 1.76 ms per flush, which matches the way-by-way walk the HAL call performs. So the flush was roughly 45% of resample, not the ~2% an earlier profile reported; that profile was taken while an experimental per-chunk flush (16x fewer flushes) was active, and the figure outlived the reverted code. What remains is bound by fabric gather throughput, not by coefficient generation — the coefficient work is hidden behind the kernel.

Resample is three workloads, and they behave differently

`sar_stage_ts` reports resample as one number, which hides the structure. `sar_resample_ts[0..3]` splits it (measured 2026-07-21, deci-1 Centerfield):

Part	Measured	Share	Shape	Parallel across lines?
Range gather	8.31 s	28%	5,634 pulse lines, Np=8192 outputs each	yes — fully independent
Corner-turn	7.33 s	25%	one global transpose of the 256 MiB frame	no — global data movement
Azimuth gather	13.53 s	46%	8,192 range-bin lines, Mp=8192 outputs each	yes — fully independent

The independence of the two gathers is provable from the host reference rather than assumed:

`resample_coeffs` is `for i in range(m): interp_coeffs(KR, kr[i])` and `apply1` is a pure per-row map, with no loop-carried dependency. The ordering between the three is strict, though — the corner-turn needs all of the range output, and azimuth needs all of the corner-turn output.

Azimuth is the largest part at 46%, and costs more per line than range (1.652 ms vs 1.475 ms) because it reads Mp=8192 samples per line against range's N=4319. Treating the two gathers as interchangeable when optimising is therefore mis-weighted; azimuth is worth about 1.6x range.

The corner-turn is the one part that does not parallelise along the same axis, and it appears **twice** in the pipeline — once here and once between the FFT passes (stage 4, also ~7.3 s) — so the two together are ~14.6 s, about 17% of the whole run, from a single shared kernel.

2.2 Window — FUSED into the range-FFT feeder (2026-07-21)

Separable Hamming taper applied as the on-the-fly product `hamr[j] · hamc[k]` from two Q15 1-D tapers, rather than a materialized 2-D table. Zero inside the zero-pad region.

There is no longer a standalone window pass. The taper is applied inside `fft_feeder_v.v` as data streams into the range FFT: `hamc` lives in a 4096×32b on-chip table (2 taps/word, so table entry n serves beat n 1:1), `hamr[row]` arrives as a scalar in the same register write that arms the row, and a 3-stage registered multiply sits on the FIFO write side. Runtime-enabled (reg 0x18 bit16) — ON for the range pass, OFF for the azimuth pass, which shares the same feeder instance.

Measured on silicon: window stage 6.0 s → 0.000 s, pipeline 87.58 → 79.79 s, and the FFT passes did not slow down (25.06 s vs ~25.07 s before), so the fused multiplies cost no wall-clock. ROI crc `0xd596c9eb` unchanged.

Two things this depends on, both deliberate:

- The arithmetic reproduces `hls_window/window.cpp` bit-for-bit, including its truncation ORDER (`cw = (hamr*hamc)>>15` first, then `(sample*cw)>>15`). That is what keeps the crc a valid gate. Note `silicon_emulator.window_fixed()` uses the OTHER order (two independent `>>15` rounds) and is therefore NOT bit-exact against the silicon it mirrors — unresolved, see §11.
- It is in hand-written Verilog, not HLS. Fusing the window into the resample gather was tried twice, was bit-exact in `shls sw` and `ll=1` in `shls hw`, and hit two distinct SmartHLS miscompiles on silicon — see `fpga/SMARTHLS_ANTIPATTERNS.md`. Do not retry that route.

2.3 The two FFT passes — a naming correction

See section 4 for the FFT engine and BFP contract.

The two FFTs are mislabelled in the code, and this doc uses the corrected names. Verified from the fixed-point mirror (`silicon_emulator.py`) and the sequencer: the frame after resample is (`Np=range`, `Mp=azimuth`), and each `fft_pass` transforms the LAST axis (columns). So:

Pipeline position	True axis it transforms	Fused into it	Historical CODE name (still in the RTL/timing printout)
FFT-1 (first pass, <code>SCRATCH→SIG</code>)	azimuth	the 2-D Hamming window	<code>rangeFFT</code> / "range FFT" / the first <code>fft_pass</code>
corner-turn	(global transpose)	—	<code>cornerturn</code>
FFT-2 (second pass, <code>SCRATCH→SIG/OUT</code>)	range	magnitude detect	<code>azFFT</code> / "azimuth FFT" / the second <code>fft_pass</code>

The code identifiers (`sar_stage_ts` labels, `SAR_SEQ_TIMEOUT_FFT1/2`, the `run_m3_iso.sh` printout strings `range-FFT/azimuth-FFT`) were NOT renamed — parsers and old logs depend on them — so when reading a raw timing dump, `range-FFT` is FFT-1 (azimuth axis) and `azimuth-FFT` is FFT-2 (range axis). This is the same swap the `sar-verification-methodology` "orientation gremlin" warns about; a transposed golden comparison is exactly what an inverted label predicts.

The corner-turn between them is the load-bearing data-movement primitive: a global transpose cannot be fused into a neighbouring stage, so it is the one stage that must fully materialize the frame in DDR. It is tiled through on-chip LSRAM and is the reason the buffer plan needs a distinct destination (`SCRATCH -> SIG` rather than in place).

Consequence for the optimisation roadmap: because **FFT-1 (azimuth axis) is fed directly by the azimuth resample pass with no corner-turn between them**, the azimuth resample can be fused into FFT-1's feeder exactly as the window already was — see the roadmap in `fpga/SAR_PIPELINE_STATUS.md`.

2.4 Detect

Per-pixel magnitude `sqrt(I² + Q²)`, saturated to uint16.

This runs IN FABRIC (since 2026-07-21), fused into the azimuth-FFT unloader: `fft_unloader_v.v` takes the magnitude as the second FFT streams to DDR, so there is no separate detect stage and no CPU involvement in the datapath at all. Cost went 19.24 s -> 0.00 s, and the azimuth FFT itself got 1.50 s faster because the unloader now writes uint16 magnitudes instead of complex int32 — half the write traffic on that pass.

It is hand-written Verilog, NOT HLS, and that is not a style preference. The HLS detect kernel was bypassed for months because SmartHLS mis-synthesized its sign extension: source-correct C (`((int16_t)(x >> 16))`) was read as UNSIGNED in the generated RTL, so every negative-I pixel overflowed and saturated to 0xFFFF — about half the image — collapsing correlation. Both `shls` cosim and a correlation check passed anyway; only a value-level comparison caught it. The Verilog declares the operands explicitly `signed`, and `tb/tb_fft_unloader_det.v` mutation-tests exactly this: stripping the `signed` qualifiers reproduces the failure at 2035/2048 mismatches.

The global block exponent forces one residual CPU pass: `emax` is not known until every row is transformed, so the unloader emits magnitudes at each row's NATIVE exponent and firmware applies `>>(emax - exp_i)` afterwards. That sweep is a uint16 shift over 128 MB — no sqrt, no sign handling — rather than the old complex-to-magnitude pass over 512 MB.

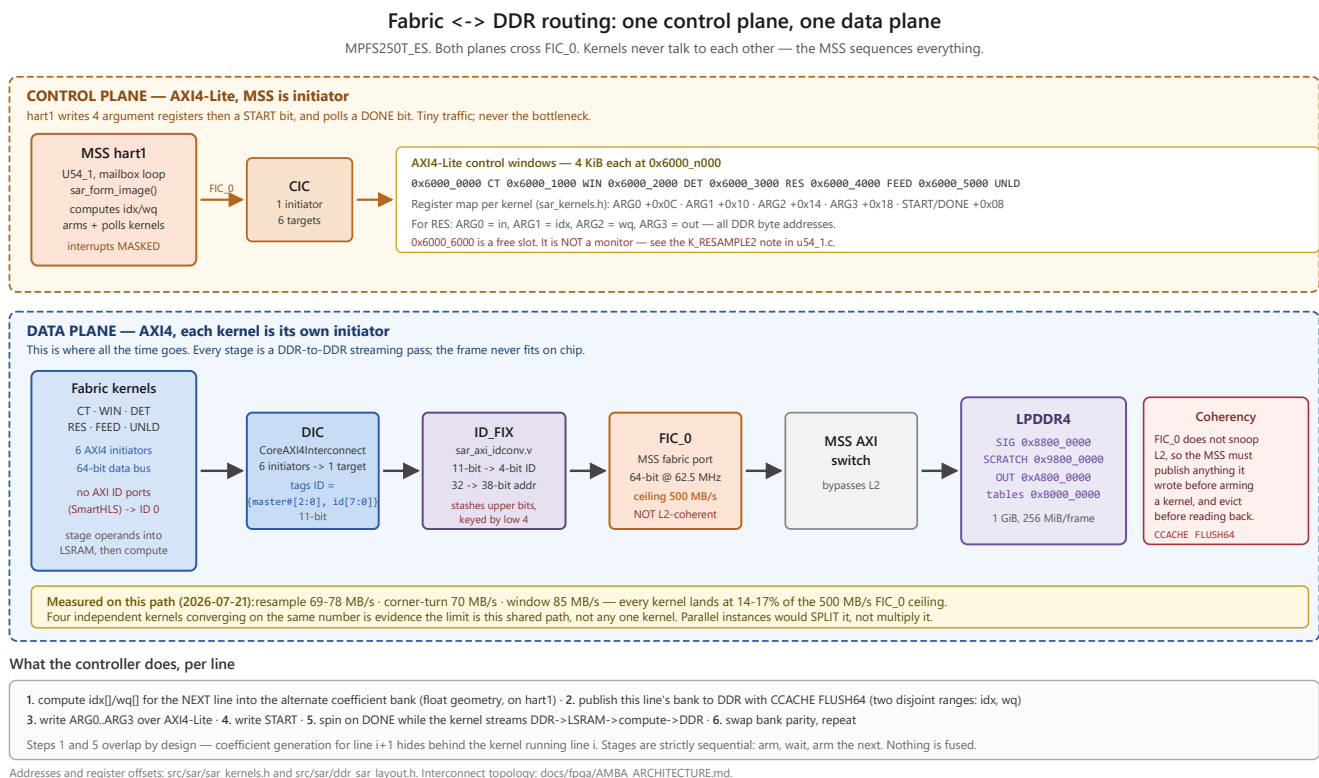
Validated by A/B against the known-good CPU detect on identical input (the pipeline CRC gate does not apply — the rounding order changes deliberately): max |diff| 2 LSB, ZERO pixels beyond that over 1,048,576, correlation 0.999866. `mpfs/host/model_detect_fusion.py` predicted that ≤ 2 LSB bound before any RTL existed.

I and Q must be read as signed int16 before squaring. The branchless reference formula is:

```
sext16(u) = (int32_t)((u & 0xFFFF) ^ 0x8000) - 0x8000
```

Engine selection is runtime: `detect_mode @ 0xB0059118`, 1 = CPU detect (the shipping path).

2.5 Data movement: how bytes get between fabric and DDR



Two planes cross FIC_0, and only one of them matters for performance.

Control plane — the MSS drives, kernels never initiate

Each fabric kernel exposes a 4 KiB AXI4-Lite target at `0x6000_n000`. The MSS reaches them through its FIC_0 initiator into a control interconnect (CIC, 1 initiator to 6 targets). Per kernel the map is `ARG0 +0x0C`, `ARG1 +0x10`, `ARG2 +0x14`, `ARG3 +0x18`, `START/DONE` at `+0x08` (`sar_kernels.h`). For resample: `ARG0 = in`, `ARG1 = idx`, `ARG2 = wq`, `ARG3 = out`, all plain DDR byte addresses.

This traffic is a handful of register writes per line and is never the bottleneck. Kernels never talk to each other and never self-sequence — the MSS arms one, waits for DONE, then arms the next.

Data plane — six initiators, one shared path

Every kernel is its own AXI4 initiator with a 64-bit data bus. All six converge on a data interconnect (DIC, 6 initiators to 1 target), then `ID_FIX`, then `FIC_0`, then the MSS AXI switch, then LPDDR4. Nothing on this path snoops L2.

The AXI ID story matters and is easy to get wrong. CoreAXI4Interconnect tags each transaction with an 11-bit ID formed as `{master_number[2:0], master_id[7:0]}`. `master_number` is the interconnect's own index of which initiator port the transaction arrived on; `master_id` is whatever the master drove. **SmartHLS `axi_initiator` kernels have no ID ports at all**, so `master_id` is always 0 and kernels differ only in the high bits — RES is `0x300`, a seventh initiator would be `0x600`.

`ID_FIX` (`sar_axi_idconv.v`) narrows that 11-bit ID to `FIC_0`'s 4 bits by forwarding the low 4 bits and stashing the upper 7 in a table **keyed by those same low 4 bits**, re-attaching them on the response. Since every kernel's low 4 bits are zero, all initiators collide on `aw_tab[0]`. That is safe only because stages run strictly one at a time — the module's header states the assumption outright: " ≤ 1 outstanding txn per distinct low-4 tag (sequential kernels)". It is a pure combinational pass-through and does **not** throttle outstanding transactions.

Consequence for any future parallelism: two kernels running concurrently would reconstruct each other's response IDs and mis-route. This is invisible to synthesis and to timing closure — such a design builds clean and fails only on silicon. Fixing it is an RTL change to `sar_axi_idconv.v` (forward `master_number` through the 4-bit tag, since `master_id` is always 0).

The measured ceiling

Stage	Bytes moved	Time	Throughput
Resample, azimuth	114,692 / line	1.652 ms	69 MB/s
Resample, range	114,692 / line	1.475 ms	78 MB/s
Corner-turn	512 MB	7.33 s	70 MB/s
Window	<i>(fused into the range FFT; no standalone pass since 2026-07-21)</i>		0.00 s —

`FIC_0` at 64-bit × 62.5 MHz has a 500 MB/s ceiling, so every kernel sits at 14–17% of it. Four independent kernels converging on the same figure is evidence that the limit is this shared path rather than any one kernel's logic — and it predicts that replicating kernels would **split** that bandwidth rather than multiply it.

What the controller actually does, per line

1. Compute `idx[]/wq[]` for the **next** line into the alternate coefficient bank (float geometry, on hart1).
2. Publish the **current** line's bank to DDR with `CCACHE FLUSH64` — two disjoint ranges, because a bank is not contiguous (section 6).
3. Write ARG0..ARG3 over AXI4-Lite.

4. Write START.
5. Spin on DONE while the kernel streams DDR → LSRAM → compute → DDR.
6. Swap bank parity and repeat.

Steps 1 and 5 overlap by design, so coefficient generation for line $i+1$ hides behind the kernel running line i . That is why coefficient work is currently "free" — and why it becomes the binding constraint the moment the gather gets substantially faster.

2.6 AXI beat packing



The gather loop, the lerp and every output value are untouched: this changes only how the same bytes are carried. Verified in the generated RTL, not inferred.

The bus is 64-bit, but **AxSIZE** decides how many of those 8 bytes a beat actually carries (2^n). The kernel originally moved **one C element per beat**: **AxSIZE 3'd2** (4 bytes) for **in**, **idx** and **out**, and **3'd1** (2 bytes) for **wq**. That is 32,769 beats per line carrying 114,692 useful bytes inside 262,152 bytes of bus time — **43.7% packing efficiency**, with **wq** wasting 75% of every beat it used.

Reading **idx** as two int32 per 64-bit word and **wq** as four int16 per word, then unpacking into the existing LSRAM arrays, cuts this to 22,529 beats and 63.6% efficiency. The gather loop, the lerp and every output value are untouched — only the transport changes.

Two streams are deliberately left unpacked, and both reasons should be re-checked before anyone "finishes the job":

- **in** — pass 1 reads **BUF_SIG + $i*N*4$** with $N=4319$, so for odd i the address is only 4-byte aligned and an 8-byte beat needs 8-byte alignment. Pass 2 would be fine, but one kernel binary serves both passes.
- **out** — producing two outputs per cycle would need four LSRAM reads per cycle from **buf** (**buf[j0]**, **buf[j0+1]**, **buf[j1]**, **buf[j1+1]**), which the two-port LSRAM cannot do; ll would go to 2 and cancel the gain.

idx and **wq** are always safe because every coefficient bank (**0xB014_8000 + $b*0x2_0000$** , and **+0x1_0000** for **wq**) is 8-byte aligned by construction.

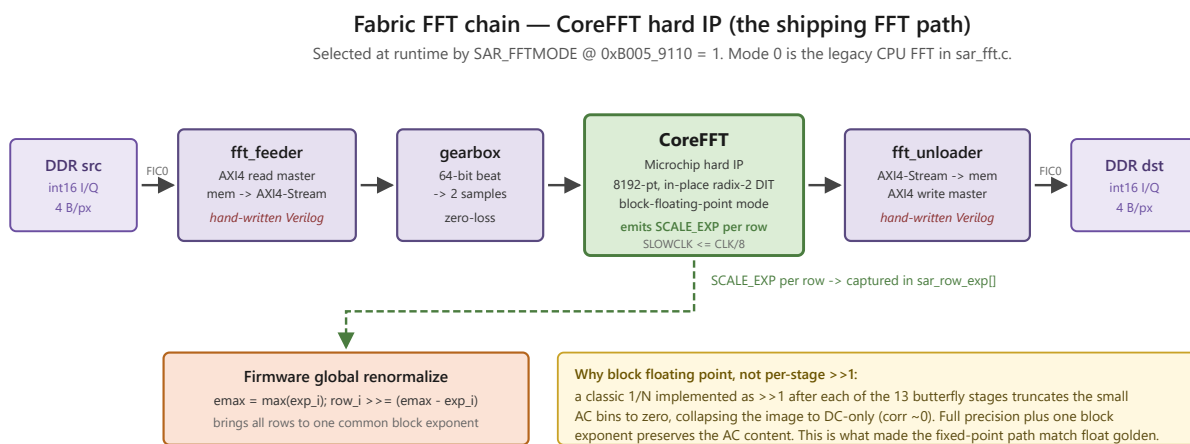
Tooling trap found doing this: a `#pragma HLS memory partition` placed anywhere other than immediately above the variable's **declaration** is silently dropped — SmartHLS warns `[HLS pragma] ignored`, exits 0, and the partitioning simply is not there. The `wq` unpack loop degraded to `ll=2` while `idx` coincidentally still made `ll=1` on the LSRAM's two native ports, so half the regression hid itself. Only the `ll` report gate caught it. See `fpga/SMARTHLS_ANTIPATTERNS.md` entry 6.

3. Fixed-point and data contracts

- Complex samples are int16 I and int16 Q packed as one 32-bit word per pixel, `hi16 = I`, `lo16 = Q`.
- The detected OUT image is uint16 magnitude, 2 bytes per pixel.
- Therefore a complex buffer is 4 B/px and OUT is 2 B/px, and **the same byte offset addresses different rows in the two**. Always compute row addresses explicitly as `base + row * GRID * bytes_per_px`. This has bitten the project more than once.
- Resample weights `wq` are Q15 (`w = wq / 32768`).
- Window tapers `hamr`, `hamc` are Q15 int16.
- Geometry arrays (`f0`, `df`, `pr`, `tans`, `krgrid`, `kcgrid`) are float32; `invorder` is int32.

Datapath sizing, from the fixed-point emulator study: 16-bit mantissa, 18-bit twiddle, 48-bit accumulate, BFP arithmetic shift after every stage. Measured FFT block-exponent growth is +9/+10 bits, covered by a 5-bit exponent per FFT line. At 16-bit the fixed image is visually identical to float (corr 0.9992 on the ship scene) and retains ~53 dB usable dynamic range at ~4.6 ENOB. If 53 dB proves marginal, 18-bit mantissa buys roughly 12 dB more and still fits one 18x18 DSP per multiply.

4. FFT engine and block floating point



Note: the IP's `SCALE_EXP` register is NOT the same quantity as a software BFP block exponent (the IP applies an unconditional 1/N scale). Do not compare the two directly.

The feeder and unloader are hand-written Verilog because SmartHLS synthesizes `mem <-> stream` kernels to dead RTL on this toolchain (`mem -> mem` HLS kernels are fine).

An earlier HLS FFT (`hls_fft_in_place_bfp`) dropped the twiddle term in generated RTL on silicon while every C-simulation passed at corr 0.9999 — it is not used. See `SAR_PIPELINE_STATUS.md`.

Re-arm hazard: the golden testbench only ran one transform. The `MEMBUF` overwrite and `READ_OUTP/DATAO`-latency traps appear only on the second and later frames.

The shipping FFT is the Microchip CoreFFT hard IP, driven as a streaming chain `fft_feeder -> gearbox -> CoreFFT -> fft_unloader`. It is selected at runtime by `SAR_FFTMODE @ 0xB0059110 = 1`, which is what the pipeline runs with; mode 0 is a retained legacy CPU FFT in `sar_fft.c`.

The feeder and unloader are hand-written Verilog. This is not a style choice: SmartHLS synthesizes `mem-to-stream` and `stream-to-mem` kernels to dead RTL on this toolchain. Memory-to-memory HLS kernels are fine.

Block floating point is the reason the fixed-point pipeline matches the float golden. CoreFFT runs in BFP mode and reports a per-row `SCALE_EXP`; the true value is `DATA0 * 2^SCALE_EXP`. Firmware captures the per-row

exponents in `sar_row_exp[]` and renormalizes globally to a common block exponent, `row_i >>= (emax - exp_i)`.

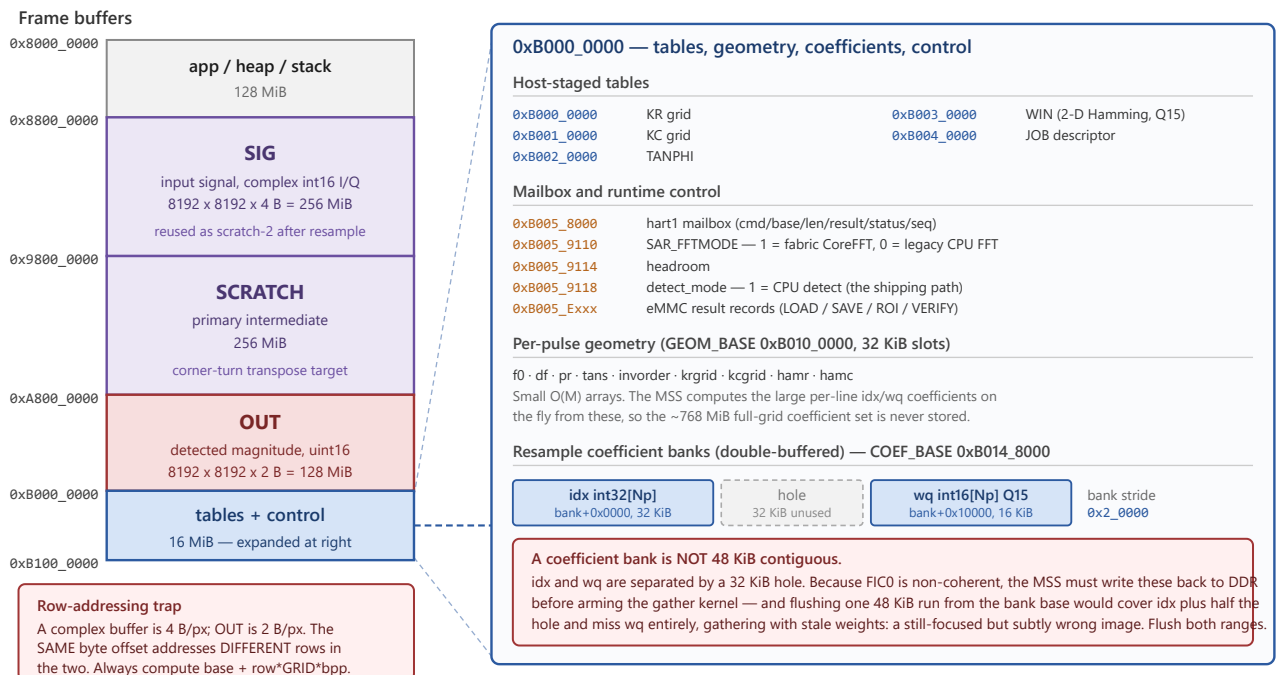
The alternative — a classic 1/N implemented as `>>1` after each of the 13 butterfly stages — truncates the small AC bins to zero and collapses the image to DC-only (corr ~0). Fixed-point FFT dynamic range must be managed with a block exponent, not per-stage truncation.

Note that the IP's `SCALE_EXP` register is not the same quantity as a software BFP block exponent (the IP applies an approximately unconditional 1/N scale). Do not compare the two directly.

5. Memory map

DDR memory map — 1 GiB LPDDR4, cached window at 0x8000_0000

Single source of truth: `src/sar/ddr_sar_layout.h`, mirrored by `mpfs/host/ddr_layout.py`. Keep the two in lock-step.



All addresses are physical, in the cached DDR window. Fabric masters reach DDR over FIC0 and do not snoop L2 — see SAR_DESIGN.md "Cache coherency" for the required flush at each handoff.

Defined in `src/sar/ddr_sar_layout.h` and mirrored by `mpfs/host/ddr_layout.py`. These two must stay in lock-step; the header is the bare-metal mirror, the Python module is the host-side single source.

Base	Size	Region
0x8000_0000	128 MiB	app / heap / stack
0x8800_0000	256 MiB	SIG — input signal, complex int16 I/Q; reused as scratch-2 after resample
0x9800_0000	256 MiB	SCRATCH — primary intermediate, corner-turn target
0xA800_0000	128 MiB	OUT — detected magnitude, uint16
0xB000_0000	16 MiB	tables, geometry, coefficient banks, mailbox and control

Buffers ping-pong SIG <-> SCRATCH so an in-place FFT never reads and writes the same page.

Within the tables region: host-staged grids at 0xB000_0000 (KR, KC, TANPHI, WIN, JOB); the hart1 mailbox at 0xB005_8000; runtime knobs at 0xB0059110/9114/9118; eMMC result records at 0xB005_Exxx; per-pulse

geometry from `GEOM_BASE 0xB010_0000` in 32 KiB slots; and the double-buffered resample coefficient banks from `COEF_BASE 0xB014_8000`, stride `0x2_0000`.

A coefficient bank is not 48 KiB contiguous: `idx` (int32[Np], 32 KiB) sits at bank+0x0000 and `wq` (int16[Np], 16 KiB) at bank+0x10000, with a 32 KiB hole between. Anything that publishes a bank must treat it as two disjoint ranges.

6. Cache coherency

FIC0 is not coherent — fabric masters reach DDR without snooping the L2. Every handoff across the CPU/fabric boundary therefore needs an explicit cache operation:

Handoff	Requirement
CPU writes data, fabric reads it	Write back L2 to DDR before arming the kernel
Fabric writes data, CPU reads it	Invalidate L2 before reading, or the CPU sees stale lines
SDMMC DMA writes DDR, CPU or fabric reads it	Write back / invalidate after the transfer completes

Two mechanisms are in use:

- `flush_l2_cache(hartid)` — the MPFS HAL's whole-cache operation. It is a way-by-way walk: for each of the 16 ways it reads 131 KiB from the L2 zero device (~268k volatile loads) and manipulates the WayMask allocation policy. Correct, but expensive, and it invalidates everything.
- `flush_coef_bank_to_ddr(bank, n)` in `sar_sequencer.c` — writes only the covering lines to the CCACHE `FLUSH64` register (`CACHE_CTRL_BASE 0x0201_0000`), which writes back and invalidates the line containing a given physical address. Used on the resample per-line critical path, where the only CPU-dirty data is the coefficient bank. Roughly 768 stores instead of ~268k loads.

Because the coefficient bank is discontinuous (section 5), `flush_coef_bank_to_ddr()` flushes `idx` and `wq` as separate ranges. A single 48 KiB run from the bank base would cover `idx` plus half the hole and miss `wq` entirely, so the kernel would gather with stale weights — producing a still-focused but subtly wrong image that a correlation check would likely not catch.

Anything a JTAG poll reads must be written back explicitly. A debugger read of DDR is a physical read: it does not snoop L2, so a value the CPU wrote but did not flush appears stale. Two cases:

- The progress word at `0xB005_9100` is published by the `SAR_PROG` macro itself, one cache line per update. This used to happen for free as a side effect of the per-line whole-L2 flush in `resample_2pass()`; once that flush was narrowed to the coefficient banks it had to become explicit. Without it the progress counter appears frozen for the whole 29.2 s resample, which is indistinguishable from a hung pipeline.
- Known gap: `MBX_CMD_CRC32`'s handler writes `mbx->result` with only a fence and no writeback, so a JTAG read of it returns a stale value. The dedicated result records at `0xB005_Exxx` are written back and are trustworthy.

A large CPU write is not self-publishing. Pass 1 zeroes the padded pulse rows — about 84 MB against a 2 MiB L2 — so most lines write-back-evict naturally as the loop advances and the region *looks* published. The final ~2 MiB stays dirty, so without an explicit flush the corner-turn reads the previous run's data in the highest pad rows instead of zeros, putting non-zero content where the FFT expects zero-padding. Whole-L2 is the right

instrument here: at 64 B per line, targeting 84 MB would be ~1.3 M **FLUSH64** stores, far worse than one way-walk, and it runs once per pipeline rather than per line.

The general rule: narrowing a flush is safe only after enumerating everything the wide flush was incidentally publishing. Cache maintenance that works by side effect is a latent dependency.

7. On-board data path (eMMC)

The scene lives on the board's soldered 8 GB eMMC, so a run needs no host data transfer. This retires the ~3 h JTAG scene load that previously preceded every run.

Two fixed-LBA partitions, each self-describing via a superblock and TOC, so job geometry is reconstructed at load time and never trusted from volatile RAM:

Region	LBA	Size	Contents
INPUT (SARI , 0x53415249)	0x80000 (256 MiB)	4 GiB	packed scene image: superblock + one blob per scene
OUTPUT (SARO , 0x5341524F)	0x880000 (4.25 GiB)	3 GiB	superblock + persisted output image(s)

A scene blob (**SARB**, 0x53415242) is a header, a segment table, and ten 512-aligned role segments: **sig**, **f0**, **df**, **pr**, **tans**, **invorder**, **krgrid**, **kcgrid**, **hamr**, **hamc**. Each segment is tagged by role and the firmware resolves role to DDR address via **sar_emmc_role_addr()**, scattering to the fixed addresses in section 5. The JOB descriptor carries only scalars (M, N, **fft_r**, **fft_a**, **out_dtype**, **bfp_in_exp**, **sig_len**, **sig_crc**) plus the SIG/OUT/SCRATCH bases — the pipeline reads geometry from the fixed addresses, not from the JOB's legacy address fields.

Measured rates (LEGACY mode, 25 MHz, 8-bit, single-block): read 1.5 MB/s, write 0.13 MB/s. Scene LOAD is 81.5 s; persisting the 128 MiB OUT image is about 16 min.

What eMMC does and does not solve: it eliminates the recurring 3 h JTAG input load, which is the win. It does not speed host transfer — dumping the full OUT image to the PC is still bound by the FlashPro6 JTAG link at roughly 9 KB/s. Keep the image on-card and dump small ROI crops to verify; a faster offload needs a different host link, not a faster card.

SDMA would speed writes by one to two orders of magnitude but is currently unusable: hart1 runs with **MSTATUS_MIE** cleared and **MSS_MMC_sdma_write** completes via the MMC PLIC ISR, so with interrupts off it spins forever in an unhaltable hang. Single-block transfers are synchronous and cannot hang.

Output persistence is commit-last: invalidate the superblock, write the image, write the superblock last. A power loss mid-image therefore leaves an invalid superblock and readers reject the torn image. Do not reorder this.

8. Boot sequence and when the pipeline runs

Nothing runs automatically. The board boots into a command loop and waits; the pipeline executes only when a host writes a **PIPE** command into the mailbox. This is deliberate — it keeps the board debuggable and makes every run explicit and repeatable.

8.1 Power-on chain

1. eNVM holds `app.elf`, programmed in boot mode 1. The MPFS HAL's startup code runs first: it trains LPDDR4 and brings up FIC_0 before any application code executes.
2. E51 (hart 0, the monitor core) enters `e51()`. It releases the MMUART0 subblock clock/reset, raises a software interrupt to hart 1, and then idles in an infinite loop forever. It does no compute.
3. U54_1 (hart 1) enters `u54_1()` and does the work:
 - spins on `MIP.MSIP` waiting for the E51 wake (an active poll, not `WFI`, so the hart stays halttable by JTAG), then clears the soft interrupt;
 - clears `MSTATUS_MIE` and `mie` — interrupts stay masked for the lifetime of the application. This is the reason eMMC SDMA is unusable (section 7): its completion path is an ISR;
 - installs a direct-mode trap vector for fault recovery;
 - clears the result table, runs the self-test battery, latches `g_m2_done`;
 - zeroes the mailbox and enters the command loop, incrementing a heartbeat each pass.
4. U54_2..4 are unused.

Boot mode matters for debuggability: mode 1 (this application) and mode 0 (a WFI stub) both leave hart 1 halttable over JTAG. An HSS build does not — JTAG then cannot halt the hart, so it is never used for this work.

8.2 What boot does *not* do

- It does not program the fabric. The bitstream is loaded separately (JTAG or SPI flash); the firmware assumes fabric is already live. A dark fabric mimics several unrelated faults — kernels never complete, and the eMMC appears dead because the eMMC/SD mux select is a fabric tie.
- It does not load a scene. DDR does not survive a power cycle, so SIG is undefined at boot.
- It does not run the pipeline.

8.3 Required order for a run

Each step depends on the previous one. Skipping one produces a failure that looks like a different problem, which is why the order is stated explicitly.

Step	Action	Why it must come first
1	Program the fabric bitstream	Kernels, and the eMMC mux select, do not exist until it is live
2	Flash the application to eNVM (<code>run_program.sh</code>)	Programming the fabric requires re-flashing the app afterwards
3	Power-cycle	The freshly flashed app must actually boot
4	<code>ELOD</code> — load the scene from eMMC (81.5 s)	DDR was wiped by step 3; SIG must be populated
5	<code>PIPE</code> — run the pipeline (48.19 s)	Needs SIG and the geometry tables in DDR
6	<code>EROI</code> / <code>ESAV</code> — crop or persist the result	Needs OUT populated

Steps 4-6 may repeat freely without re-flashing, provided the board is not power-cycled between them. A power cycle at any point returns you to step 4, never earlier.

8.4 Reading state over JTAG

How a debugger reaches memory changes what it sees, which explains an otherwise confusing inconsistency:

- While the hart is halted, accesses go through its progbuf — they use the hart's own load/store path and are therefore cache-coherent. This is why the host can write the mailbox and poll `mbx->status` without any flush.
- While the hart is running, the debugger must use the system bus, which is a physical read that does not snoop L2. Anything polled during a run — notably the progress word at `0xB005_9100` — must be explicitly written back by the firmware, or it appears frozen.

9. Control interface

The hart1 mailbox at `0xB005_8000` is the single entry point. Layout: `cmd` +0, `base` +4, `len` +8, `result` +0xC, `status` +0x10 (done = `0xC0FFEE03`), `seq` +0x14. Write `base` and `len` first and `cmd` last; the hart clears `cmd` to acknowledge.

Command	Code	Purpose
ELOD	0x454C4F44	Load scene eMMC -> DDR, plus JOB
PIPE	0x50495045	Run the image-formation pipeline
ESAV	0x45534156	Persist OUT -> eMMC SARO
EVOU	0x45564F55	Verify the persisted image by full CRC against the TOC
EROI	0x45524F49	Crop an ROI from the DDR OUT image
EROE	0x45524F45	Crop an ROI from the persisted SARO image
EPRV	0x45505256	Provision the INPUT partition
EMMC	0x454D4D43	eMMC self-test

Results are latched in dedicated DDR records (`0xB005_E000` LOAD, `E100` SAVE, `E200` ROI, `E300` VERIFY, `0xB005_D000` provision), each written back to DDR so a JTAG physical read sees them. Verdict 0 is pass; 1 PARAM, 2 INIT, 3 MAGIC, 4 IO, 5 CRC.

Per-stage timing is instrumented in `sar_stage_ts[0..6]` from MTIME at 1 μ s per tick, readable without re-running the pipeline via `bash mpfs/host/run_stage_timing.sh`.

Fabric kernels are controlled over AXI4-Lite through FIC0; per-kernel register offsets are in `src/sar/sar_kernels.h`. Note there are two register-map models in the project's history — the hardware uses the per-kernel model in `sar_kernels.h`, not the older monolithic one.

10. Verification contract

Correlation is scale-, phase- and orientation-invariant, so it hides real bugs — the detect sign-extension defect passed a correlation check while corrupting half the image. Verify by value: feed known inputs, diff actual complex sample values against the bit-accurate emulator (`silicon_emulator.py`, which equals the float golden at corr 1.0), and only then compare to the reference image.

When comparing a board image to the golden, account for orientation before declaring a divergence. The board result matches the golden in the `T.rot180` orientation (`board == golden.T[::-1,::-1]`). A naive band comparison once read corr 0.06 on a correct image that scanned to 0.97 — run the full 8-dihedral orientation search (`mpfs/host/correlate_cpufft.py`) first.

Current reference result: corr 0.9923 against `golden_small_mag.npy` on the Centerfield decimated scene, with a point-target crop at 0.9962.

11. Known deviations from the ideal design

These are deliberate and load-bearing. Do not "fix" them without reading the linked history.

Deviation	Reason
Detect is fabric, in the FFT unloader (not a kernel of its own)	The standalone HLS detect kernel is unusable (sign-extension miscompile). Fusing it into <code>fft_unloader_v.v</code> in Verilog both fixes that and deletes the 19.24 s pass (§2.4)
Resample gather is <code>II=1</code> but ~2.44x AXI-stalled	The kernel schedules at <code>II=1</code> on all 4 loops (22,545 cyc = 361 us/line) yet measures ~880 us -- AXI stall on a correct schedule, not a burst failure (an earlier "single-beat reads" claim was a stale-report error). Localising it needs the <code>FIC_0 ARLEN+gap</code> monitor. Largest remaining target
FFT feeder/unloader are hand-written Verilog	SmartHLS mem <-> stream kernels synthesize to dead RTL
Window is fused into the FFT feeder (Verilog), not into resample (HLS)	Two distinct SmartHLS miscompiles on the resample-fusion route; the Verilog feeder route works and is silicon-proven (§2.2)
<code>silicon_emulator.window_fixed()</code> is NOT bit-exact vs <code>window.cpp</code>	It applies the two tapers as two separate <code>>>15</code> rounds; the kernel folds them into one <code>cw</code> first. Differs in the low bit. Pre-existing, found 2026-07-21, unresolved — the mirror's docstring claims bit-accuracy, so one of the two should change
eMMC uses single-block transfers, not SDMA	Interrupts are off on hart1; SDMA would hang unhaltable
Fabric runs at 62.5 MHz	75 MHz effort stopped at marginal ROI; the HLS window/resample kernels are the Fmax limiter, not CoreFFT or the DDR bridge
~50% of OUT saturates at 65535	Detect BFP shift headroom; cosmetic, correlation is measured on unsaturated pixels

SmartHLS is treated as an untrusted, behavioural-only tool. Every kernel output is value-checked on silicon after a rebuild — see the `hls-trust-harness` skill and `fpga/SMARTHLS_ANTIPATTERNS.md`.